JOnAS
Java Open Application Server

The ObjectWeb J2EE™ Platform and Apache Projects

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Trent Shue
Trent.Shue@bull.com





Agenda

- ObjectWeb Consortium
- Brief Overview of J2EE™
- JOnAS
- Apache Projects within JOnAS
- Other Open Source Projects from ObjectWeb within JOnAS
- ObjectWeb/JOnAS Apache/Geronimo Synergy

The ObjectWeb Consortium

Open Source Middleware



ObjectWeb Consortium: Description and Goals

Open Source Initiative

- Founded in 1999 by France Telecom R&D, Bull, INRIA

Develop Open Source Middleware

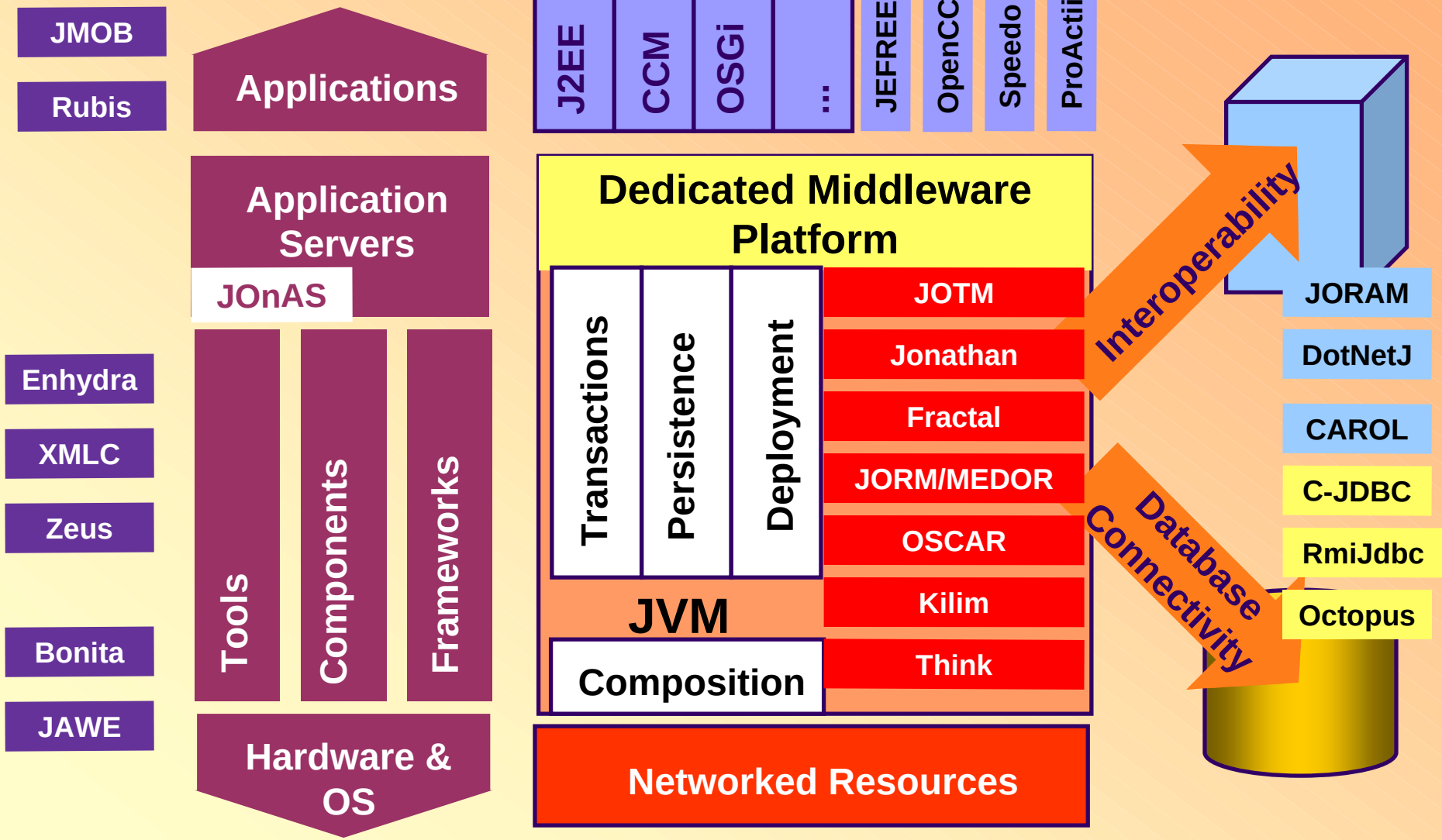
- Distributed component-based middleware technology, based on industry standards
  - CORBA: CORBA, CCM, etc.
  - Java: J2EE/EJB, JDO, JDBC, etc.
  - W3C: HTTP, SOAP, etc.
- Software available under LGPL

International Consortium

- Hosted by INRIA
- Since February 2002

An Architecture for Component Based Development

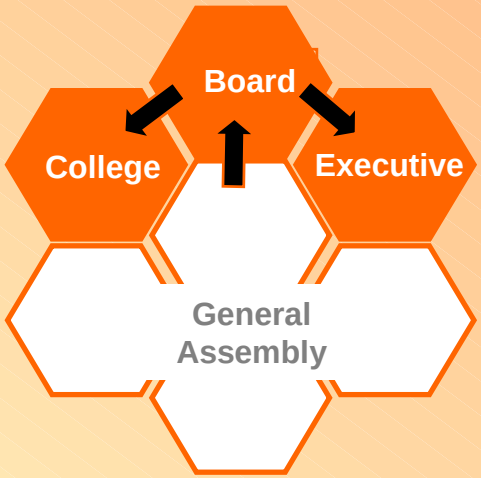
ObjectWeb
Open Source Middleware



ObjectWeb Consortium

With Open Source Principles

- **Collaborative & Reuse**
  - Experts & Professional Ecosystem
  - Collaborative Infrastructure
- **Independence & Control**
  - Open Source Components & Open Standards
  - Self Conducted Projects
- **Innovative & Self Sustainable Software Economy**
  - Seeding from Research
  - “Free” Training Material



Industrial Open Source Middleware

- **For Mission Critical & Distributed Applications**
- **Initiated In Europe**
  - with Industry Players
- **LGPL Software**
  - Frameworks, libraries, servers, tools
- **Compliant with open standards J2EE, XML, CORBA, OSGi**

A Dedicated Organization

- **Projects Valorization**
  - Board of Directors (strategy)
  - Executive Committee (development)
- **Technical Vision Development**
  - College of Architects
- **Basic Principles**
  - Collegiality
  - Meritocracy
  - Active Participation

Toward Success

- **ObjectWeb**
  - >400 members
  - >45,000 visitor per month
  - >1,200,000 hits per month
  - 50 « active » projects
  - 140 developers
  - >2000 users
- **JOnAS**
  - >150,000 JOnAS downloads
  - 500 users on the JOnAS list
  - Embedded in Mandrake & Red Hat (soon) distributions
  - J2EE 1.4 Certification Scholarship granted by Sun

Brief Overview of J2EE™



Java™2 Enterprise Edition

"A set of coordinated specifications and practices that together enable solutions for developing, deploying, and managing multi-tier server-centric applications."

Java™ 2 Enterprise Edition
Goals

- **Component based, interoperable applications**
- **Services-oriented architecture providing**
  - *High availability, Scalability, Reliability, Security*
- **Application life-cycle support**
  - Component development, application assembly, deployment, management
- **Application Portability**
  - Operating System and Application Server
- **Declarative Configuration**
  - Insolating the application developer from systems details

Java™ 2 Enterprise Edition

What is an Application Server?

An Application Server provides the business logic for an application program. It serves Web requests, manages database access, security, transaction services, etc…

Building Blocks of J2EE™

Components, Containers and Connectors

The J2EE™ application model can be divided into three fundamental parts:

- **Components**

  Components are the application programmer's responsibility.

  There are two types of components:

  - **User interface – (Java, Applet, Servlets, JSPs),**
  - **Business logic - Enterprise Java Bean framework.**



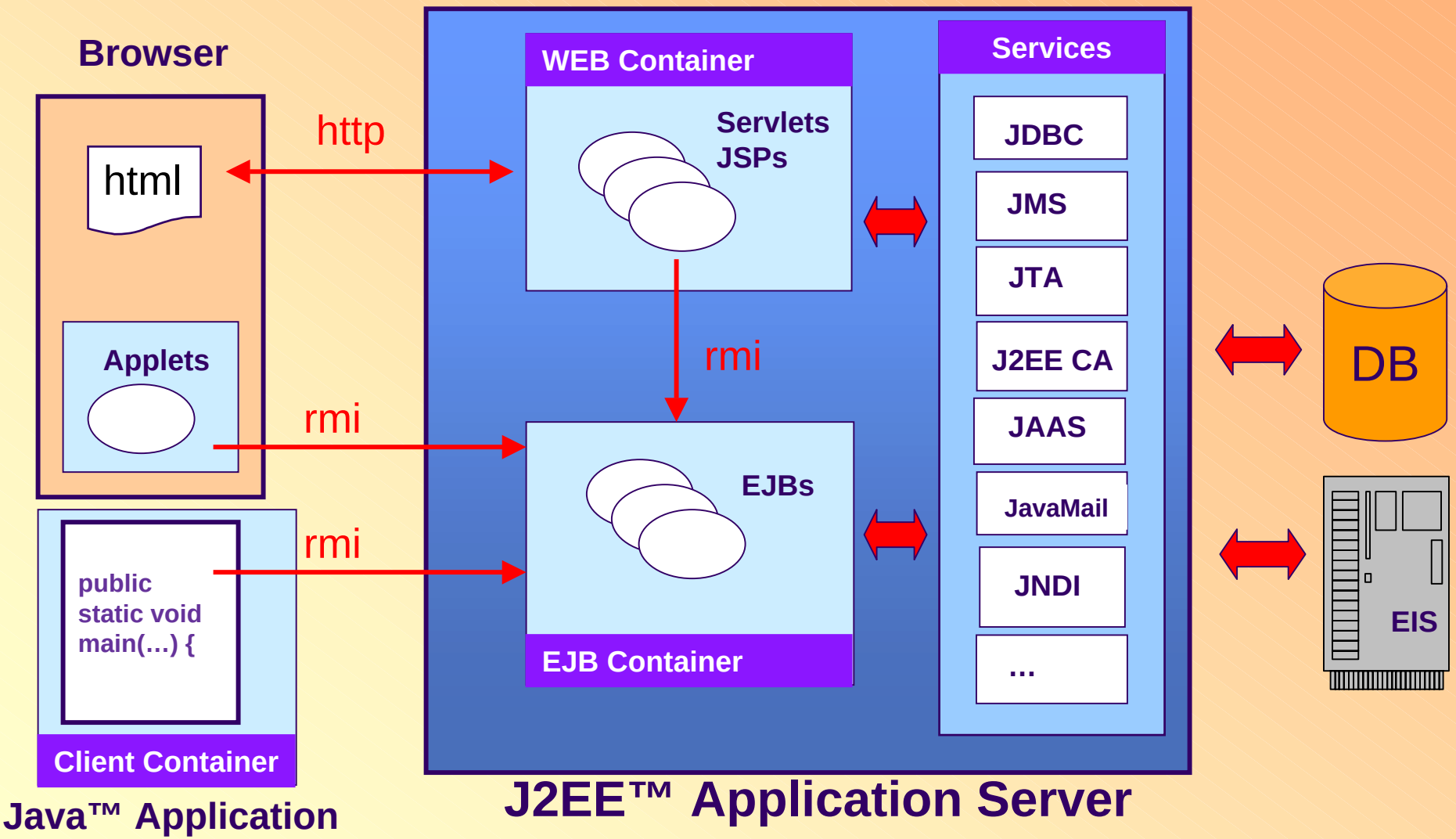
- **Containers**

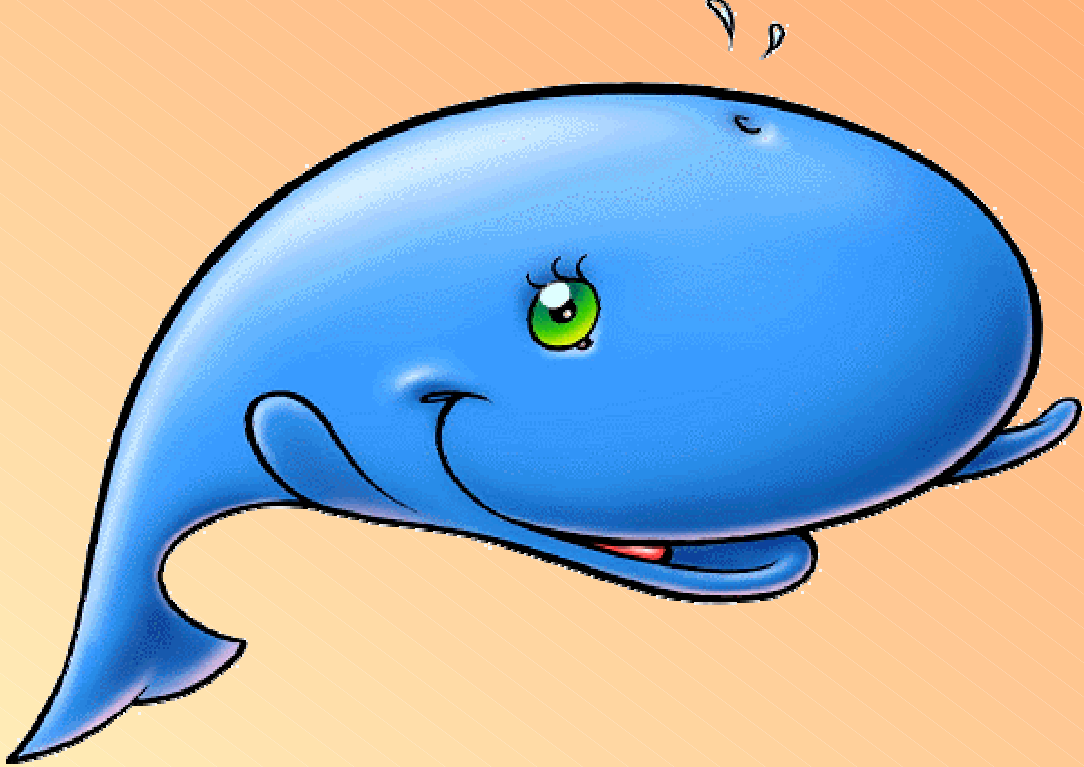
  Systems-level run-time hosting structures for J2EE™ components, providing the interfaces between the components and the application server and services. Containers provide components with services such as lifecycle management, security, deployment, and threading.

- **Services**

  Services are systems-level software components made available to enterprise application components through standardized contracts. Examples include: Resource service for access to EIS data, Transaction service for access to transaction managers, JNDI – The Java Naming and Directory Interface service, etc.

J2EE™ Architecture





The JOnAS J2EE™ Platform

Java Open Application Server

ObjectWeb
Open Source Middleware

www.objectweb.org

JOnAS Objectives

- **J2EE platform for Component based development**
  - develop presentation and business logic components, platform independent, that will support *declarative* and *configurable* transactional, persistence and security behaviors.
- **JOnAS Strengths**
  - Scalability
  - Management facilities
  - JMS, J2EE™CA, JTA support
  - Services based architecture
  - Distribution (multi-protocols: jeremie, jrmp, iiop)
  - Clustering (partially available)
  - Web Services (J2EE 1.4)

JOnAS
Availability and Requirements

- **JOnAS is distributed under LGPL (GNU Lesser General Public License)**
- **Available at:**
  - **jonas.objectweb.org**
    - **tgz or exe (windows installer) packages:**
      - **JOnAS binary or source**
      - **JOnAS-Tomcat-Axis**
      - **JOnAS-Jetty-Axis**
- **Requirements**
  - JDK1.3 or 1.4 (Sun, IBM, BEA JVMs)
- **Supported Environment**
  - Operating Systems : Solaris, AIX, Windows, Linux, HP-UX, etc.
  - Databases : PostgreSQL, MySQL, Oracle, SQL Server, DB2, Access, Objectstore, Interbase, InstantDB, Sybase, Informix, Versant, etc.

JOnAS
Java™ Compliance

EJB Container

EJB 2.0

EJB 1.1

WEB Container

Servlet 2.3
JSP 1.2 *Tomcat 4*



J2EE™ 1.3

- **JCA 1.0**
- **JDBC 2.0**
- **JTA 1.0.1**
- **JMS 1.0.2 / *JMS 1.1*** *Joram 3.4*
- **JMX 1.1** *MX4J or Sun RI*
- **JNDI 1.2.1**
- **JavaMail 1.3**
- **JAAS 1.0**
- ***JAX-RPC 1.0, JAXM 1.1, JAXR 1.0, JAXP 1.2***

JOnAS
Features [1/2]

- **JMX based management**
  - Web GUI (Struts)
- **Web services**
  - AXIS integration
  - Web Services Endpoints and Clients deployment
- **Clustering**
- **J2EE Connector architecture support**
  - possibility to deploy J2EE CA compliant Resource Adapters
- **JOnAS services**
  - possibility to launch services to be run together with the EJB server, and to be used by EJB applications. Most of the components of the JOnAS EJB server are pre-defined JOnAS services (TM, JMS, JMX, ...).
- **Scalability/Optimizations**
  - Pooling (connections, instances, threads)
  - Data access optimizations (cache, …)

JOnAS
Features [2/2]

- **Interoperability**
  - Through Web Services
  - Between Application servers (RMI/IIOP)
  - With CORBA
- **Multi-protocol support**
  - Through CAROL ObjectWeb project
  - RMI/JRMP, RMI/IIOP, JEREMIE, CMI
- **Enhanced configuration and deployment facilities**
  - ejb-jars, wars, ears directories (autoload directories)
  - Deployment from the admin console or automatic
  - JONAS_BASE: possibility to define several application environments (for configuration and deployment)
- **Security**
  - EJB permissions management
  - JAAS LoginModules support
  - Certificate based authentification

JOnAS Architecture

ObjectWeb
Open Source Middleware

HTML Clients





RMI Clients

Appli.ear

Apache

HTTP

EJB Container

EJBs

WEB Container

JSPs
Servlets

Communication
Messaging
EJB
Transaction
Security
EAR
JCA
HooX
Management
Web Container
JavaMail
Web Services
Database

DB

JDBC Calls

JORAM

JOTM

CAROL/Jonathan

Services

Tomcat / Jetty

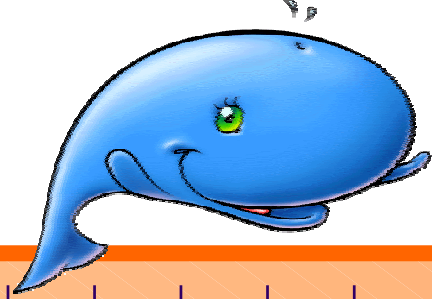
AXIS

JORM / MEDOR

GCOS

JOnAS J2EE Server

JOnAS Roadmap



Nov2 002
Jan 2003
une 003
Dec 2003
Jan 2004
June 2004

JOnAS 2.6.1

JOnAS J2EE Platform (1.3-)

J2EE packagings
AXIS
Enhanced Mgt 1
JONAS_BASE

Full J2EE 1.3
Client Container
Clustering EJB
Enhanced Mgt 2
Security
Dev tools integ
CAROL

Cluster/domain mgt
EJB level failover
...

JOnAS 3.2

JOnAS 3.0

JOnAS 3.1

JOnAS 3 /Itanium

J2EE 1.3 (CMP2.0)
Clustering WEB
JCA enhancement
Mandrake 9.1 dist

JOnAS 3.3

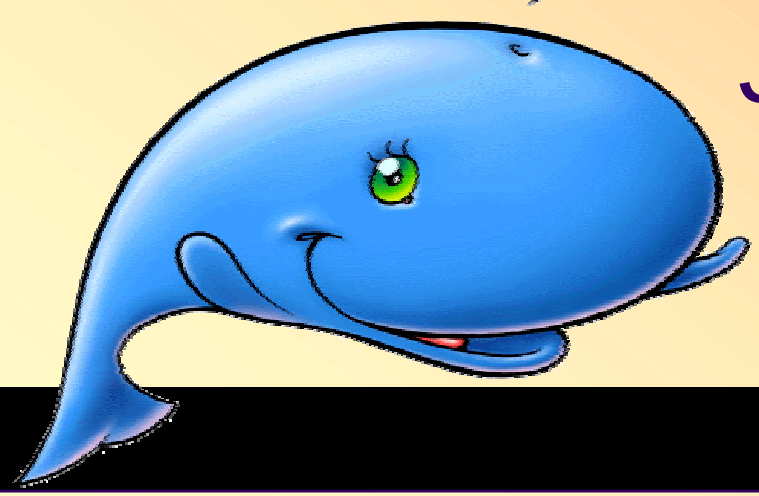
JOnAS 4

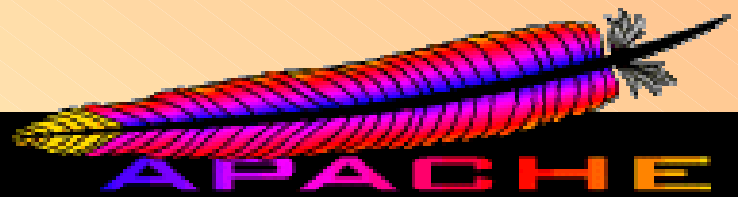
JOnAS 4.x

J2EE 1.4 Web Services
EJB 2.1 Timer Service
Certificate auth
JDBC RA
C-JDBC
New mgt console
completed

Full J2EE 1.4
Tomcat 5
J2EE depl API&tool

APACHE Projects within JOnAS





ObjectWeb
Open Source Middleware

Detailed List

Apache Project	Version	Used for
APACHE	2	HTTP server/clustering
ANT	1.5.4	Build tool
AXIS	1.1	Web Services engine
Digester (*)	1.5	XML parsing
Log4j	1.2.8	Logging
mod_jk		HTTP Clustering
STRUTS	1.0.2	JonasAdmin console
Tomcat	4.1.27	Web container
Velocity	1.2	Template based code generation (GenIC, Newbean)
Xalan	2.5.1	HTML formatting of On-line doc and tests results
Xerces	2.5.0	XML parsing

(*) and related commons packages: beanutils, collections, logging, modeler

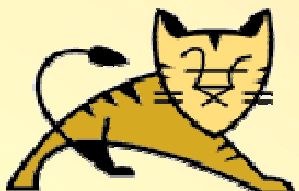
Tomcat as a Web Container



- **Servlet/JSP engine: Tomcat 4 for J2EE 1.3, Tomcat 5 for J2EE 1.4**
- **JOnAS add-ons**
  - JOnAS provides J2EE context support (java:comp/env), uniform naming of Tomcat deactivated
  - Security realms enhancements in order to propagate security contexts to EJBs
  - Add of JonasStandardContext for pre-configured contexts in server.xml (instead of creating a new one at war deployment)
  - Configurable delegation model for classloaders: either Java2 (ask parent classloader first, default) or usual web container (ask current classloader first)
  - Specify context root and host names of a war in jonas-web.xml (may also be done in a JonasStandardContext)
- **jonas-tomcat packaging available**

How JOnAS was inspired by Tomcat

- **Configuration scheme:**
  - JONAS_BASE similar to CATALINA_BASE
  - Separate configuration instances from installation)
- **The JOnAS Management Console**
  - Inspired from the Tomcat (Struts based)
- **Deployment of EARS**
  - Unpackaged directory as Tomcat does for Wars
- **Jakarta Commons Digester for XML parsing within JOnAS,**
  - As in Tomcat 5 (for dealing with both DTDs and XML schemas).



ANT



- **The complete build process of JOnAS is based on ANT**
- **J2EE applications may be compiled and deployed for JOnAS using ANT**
  - JOnAS specific nested element of the ANT ejb-jar task
  - Further tasks for starting/stopping the JOnAS server under study
- **JOnAS Tests result forms generated through the JUnitReport ANT task and the Apache XSLT stylesheet processor Xalan**

Web Services Support through AXIS [1/2]

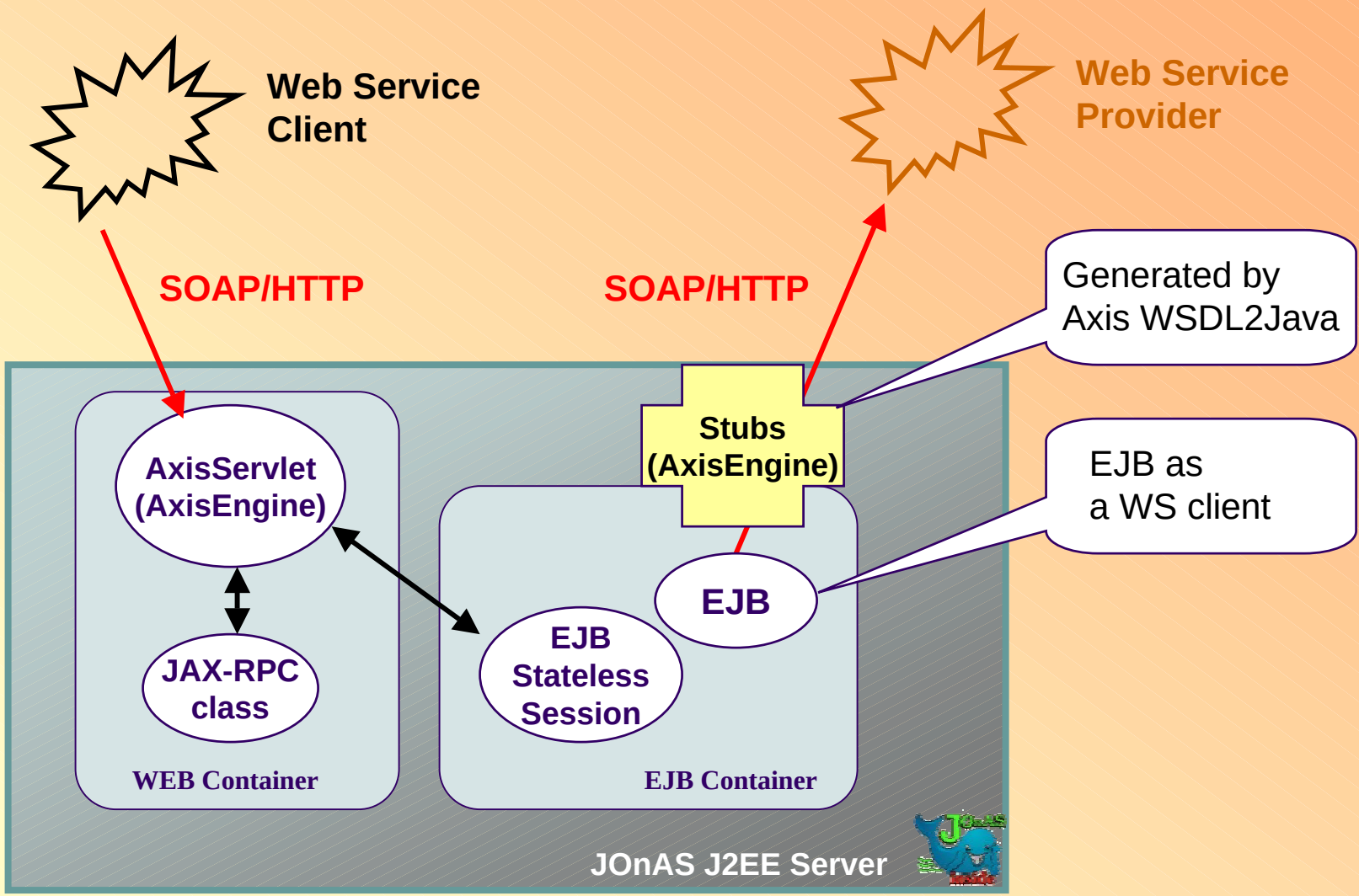
J2EE 1.4 Requirements

- Web Services Endpoints
  - « EJB session stateless » or « JAX-RPC class in the Web container » may be exposed as a Web Service
- Web Services client
  - Any J2EE component (Servlet/JSP/EJB) may access external Web Services

AXIS Support

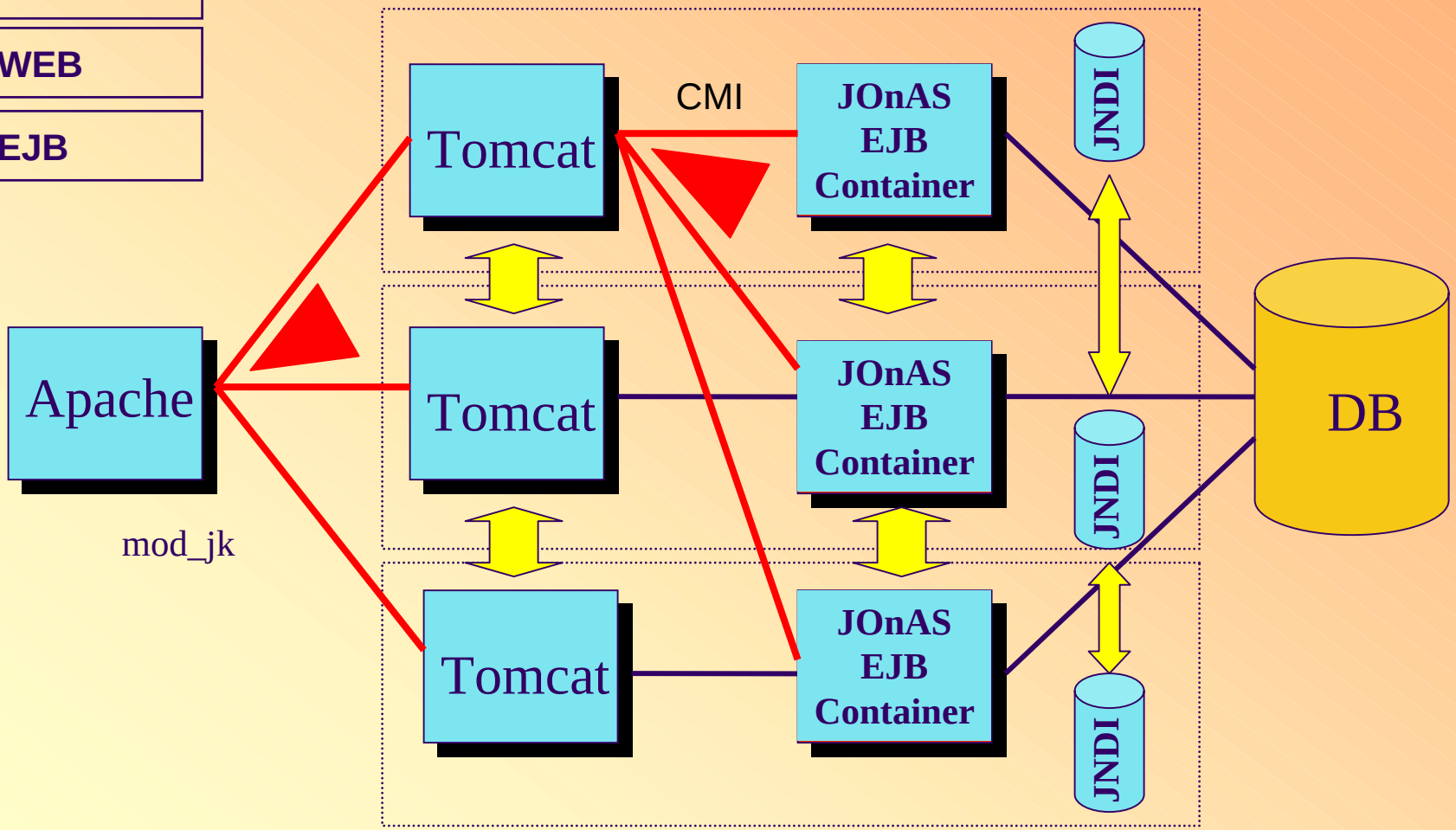
- AxisServlet in JOnAS Web Container for providing Web Services Endpoints
- Axis deployment/generation tools used by JOnAS
  - Use of WSDL2Java for generating Client stubs
  - Use of Axis configuration files (server-config.wsdd, client-config.wsdd), will be generated in a near future

Web Services Support through AXIS [2/2]



Apache and mod_jk for JOnAS Clustering

- **LB WEB**
- **LB EJB + clust JNDI**
- **Failover WEB**
- **Failover EJB**



JonasAdmin:
STRUTS-based Management Console [1/3]

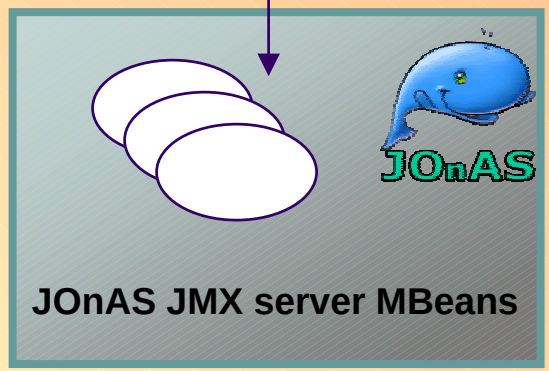


STRUTS Controler Servlet

STRUTS Actions

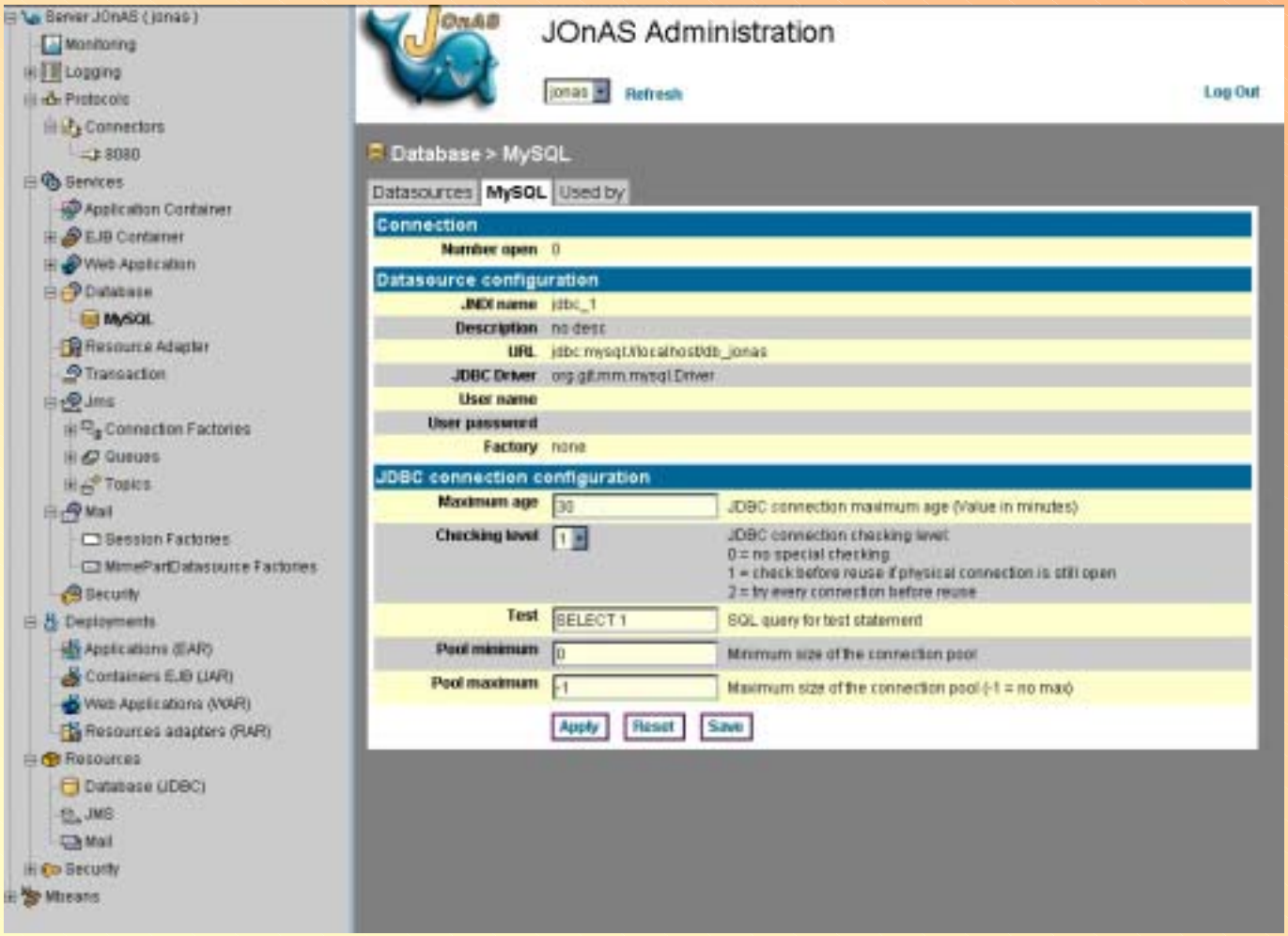
ActionForm Bean

JSP

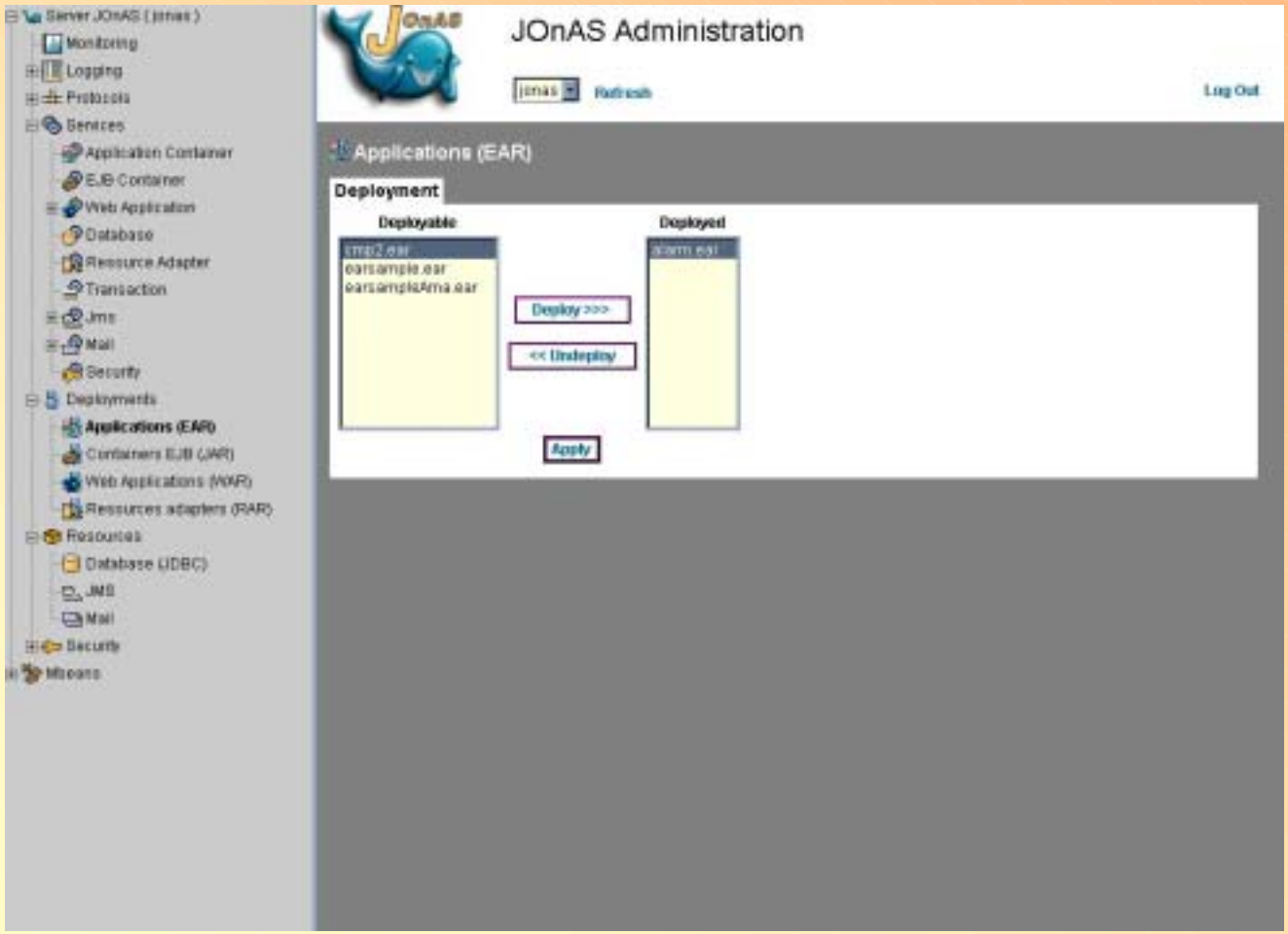


+ JOnAS specific tag lib for JSPs

JonasAdmin: STRUTS-based Management Console [2/3]



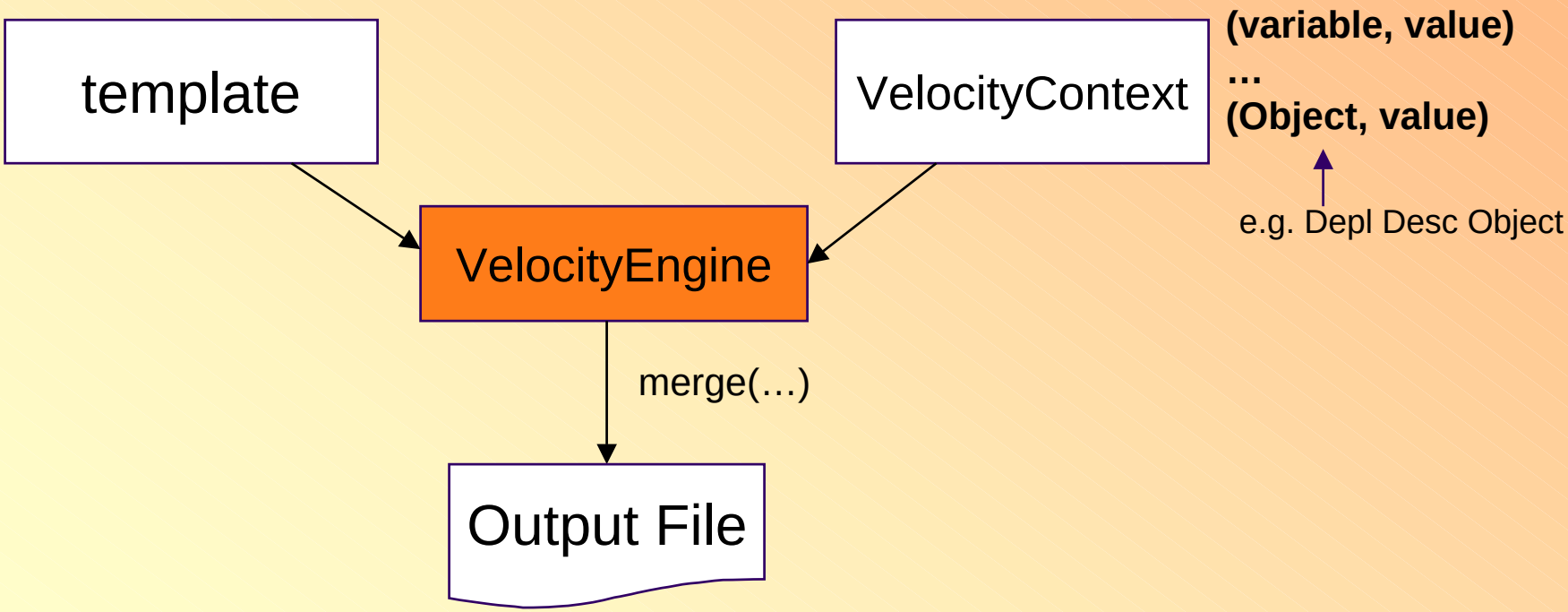
JonasAdmin: STRUTS-based Management Console [3/3]



Velocity

Template based code generation tool

- Used for JOnAS interposition classes generation (deployment tool GenIC)
- Used for EJB skeletons generation within the EJB Wizard tool NewBean delivered with JOnAS



Digester

- **Used by JOnAS 4 for parsing XML deployment descriptor files**
- **Will also be used for XML configuration files**
- **Why Digester: Flexibility**
  - Handles both XML schemas and DTDs
  - Handles incompatible XML grammars
    - e.g. EJB 2.0 DTD and EJB 2.1 schema are incompatible, but an Application Server should be able to deal with both of them
    - Impossibile to rely on tools statically generating classes from a schema, like Enhydra/Zeus, Castor or JAXB
    - This is made possible by the use of Digester's programmatically defined rules

Other Open Source Projects from ObjectWeb



JORAM

Java™ Open Reliable Asynchronous Messaging

- JORAM is a pure Java implementation of JMS 1.1
- JORAM is a free, open source initiative from ObjectWeb.
- Provides access to a Message Oriented Middleware
- Distributed Architecture improves flexibility, availability, and scalability
- Integrated with JOnAS EJB platform
- KJoram: a lightweight « client » side access from portable devices (J2ME)
- Use of TCP or SOAP protocol for JMS client to JMS server communication
- Supports Point-to-Point and Publish/Subscribe Domains
- Hierarchical topics
- "Clusterizable" topics
- Will eventually use the J2EE™ Connector Architecture

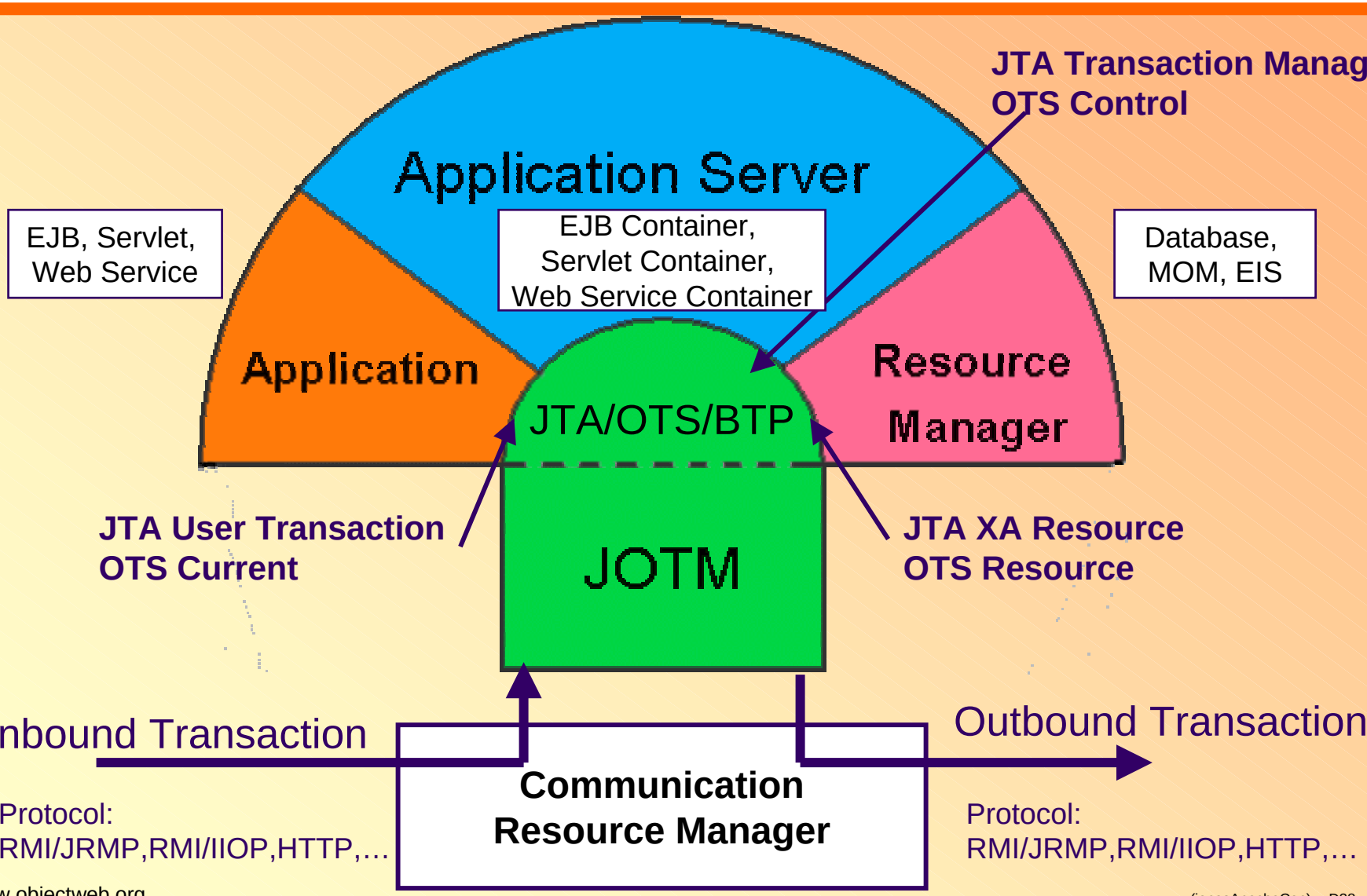


Java Open Transaction Manager

- **Supports several Transaction Models/Specs**
  - J2EE™, CORBA, Web Services
- **Implements the Java Transaction API (JTA)**
  - RMI/JRMP and RMI/IIOP Protocol support
  - Implements Web Services
    - Business Transaction Protocol (BTP)
- **Full Distributed Transaction Support for Java applications**
- **JOTM is bundled with JOnAS, JETTY and OFBiz**

ObjectWeb
Open Source Middleware

JOTM
Java Open
Transaction Manager



ObjectWeb
Open Source Middleware

JORM

Java Object Repository Mapping

- Adaptable persistence service
- Provides object persistency through different secondary storage mechanisms
  - **Files**
  - **Relational Databases**
  - **Object-oriented Databases**
- Various personalities
  - **CMP EJB™ specification**
  - **OMG PSS specification**
  - **JDO™ (Java Data Objects) specification**

MEDOR Middleware Enabling Distributed Object Requests

- MEDOR is a query framework
- The MEDOR project allows the expression, optimization and evaluation of queries on heterogeneous distributed objects.
- It is used for building middleware which require query capacities.
- MEDOR personalities include EJB 2.0, with the JOnAS EJB server, and JDO with the Speedo implementation.
- MEDOR is integrated with the persistence framework JORM.

Common Architecture for RMI ObjectWeb Layer

- CAROL is a library allowing use of multiple RMI implementations.
- Java server application can be independent of RMI implementations and accessible simultaneously by clients using different RMI implementations.
- CAROL allows the design, implementation, compilation, packaging, deployment and execution of distributed applications compliant with the RMI model.

ObjectWeb/JOnAS
Apache/Geronimo Synergy



ObjectWeb
Open Source Middleware

ObjectWeb & Apache

- Apache & ObjectWeb are loyal to
  - **Open Source rules** (NO RESTRICTION)
  - **Open Standards** such as J2EE
- Apache & ObjectWeb are true **non-profit, eligible for certification**
- competition is good**!**
- **share ideas and code**, to standardize tools and approaches (reusability, component model, etc.)

- **Opportunity for technical synergy**
  - ObjectWeb and Apache **complement** each other in their components offering
  - Objectweb's teams are positive about **sharing code** with Apache knowing that most of **ObjectWeb's projects** are **using** Apache/Jakarta components

ObjectWeb - Apache Collaboration

- **ObjectWeb components within Geronimo ?**
  - for ObjectWeb components that are to be developed collaboratively with the Apache Software Foundation, a change of license from LGPL to BSD is encouraged by ObjectWeb Board and College (decision under responsability of source code copyright holders)
- **Two first candidate components**
  - JOTM (Java Open Transaction Manager)
  - ASM (Java bytecode manipulator)
  - *Copyright holders are ready to make the license change, should the ASF require it*
- **Component sharing issues**
  - ObjectWeb provides a component model (FRACTAL) leveraging reusability, interchangeability, configurability of software components
  - Apache Avalon project
  - *Cooperate on a component-based middleware approach !*

Appendices



Appendix : JOnAS vs JBOSS

Technical features

- Both support EJB 2.0 and J2EE 1.3
- JOnAS provides a true Web management interface (the Jboss one is a JMX Mbeans generic browser)
- JOnAS integrates with high quality components:
  - Both Tomcat and Jetty, better transaction management (JOTM), highly distributed and scalable messaging service (JORAM), persistence framework (JORM)
- JOnAS will soon provide flexible and generic transaction management (JTA, JTS, BTP, …): through JOTM
- Both have an optimized communication layer. Better performance and scalability for JOnAS (See http://www.cs.rice.edu/CS/Systems/DynaServer/perf_scalability_ejb.pdf)
- Ease of use, free and up to date documentation for JOnAS

Community

- Scope: adaptable middleware versus restricted to J2EE
- Objectives: community objectives versus corporate objectives
- Industrial quality middleware components

Some JOnAS industrial uses

- **Quipsy (D)** www.quipsy.de
  - Software for quality in manufactoring or service companies
- **Deutsche Post (D)**
  - Infrastructure management (19 servers and 200 users)
- **High Sierra Technology (Sp)** www.hisitech.com
  - Failure and repair management for a transport company
- **Ingenosya (F)** – www.ingenosya.com
  - ANJARY: Hotel management
- **Intelligent Computer Systems (USA)** – www.icsaward.com
  - e-commerce, tracking / inventory systems, order entry
- **Kovair (USA)** – www.kovair.com
  - B2B relationship Management
- **OpenUSS (D)** – openuss.sourceforge.net/openuss
  - University Portal (10 000 users)
- **SportMaschine (D)** www.sportmaschine.de
  - Content Management System for Sport clubs and organizations
- **OpenWide/Thales (F)**
  - French parking fine payment system

Some J2EE™ offerings

- **Commercial**
  - BEA WebLogic
  - IBM Websphere
  - Sun One
  - Borland Enterprise Server
  - Oracle 9i Application Server
  - Macromedia jRun
  - Iona Orbix E2A
- **Open Source**
  - Geronimo (under development)
  - JBOSS
  - JOnAS
  - OpenEJB

Business Services
Enterprise Java Beans (EJBs)

Java components that implement business logic.

- **Session Beans:** Transient objects, attached to a client.
  - **Stateful** = Conversational – State is retained for the life of the session.
  - **Stateless** = State is retained for the life a a method invocation.
- **Entity Beans:** Persistent objects, implementing the use of a storage mechanism for persisting state, usually a database.
  - **Bean Managed Persistence** – Persistence is implemented in the bean.
  - **Container-Managed Persistence** – Persisting done by the container.
- **Message-driven Beans:** Asynchronous objects, executing on receipt of a JMS message.

Presentation Services

Servlets and JSPs

- **Servlets**
  - Java component for generating HTML pages
- **Java Server Pages (JSPs)**
  - HTML pages with embedded Java code



Dynamic HTML pages may access resources of the J2EE™ Application Server, mainly the EJBs.

EJB

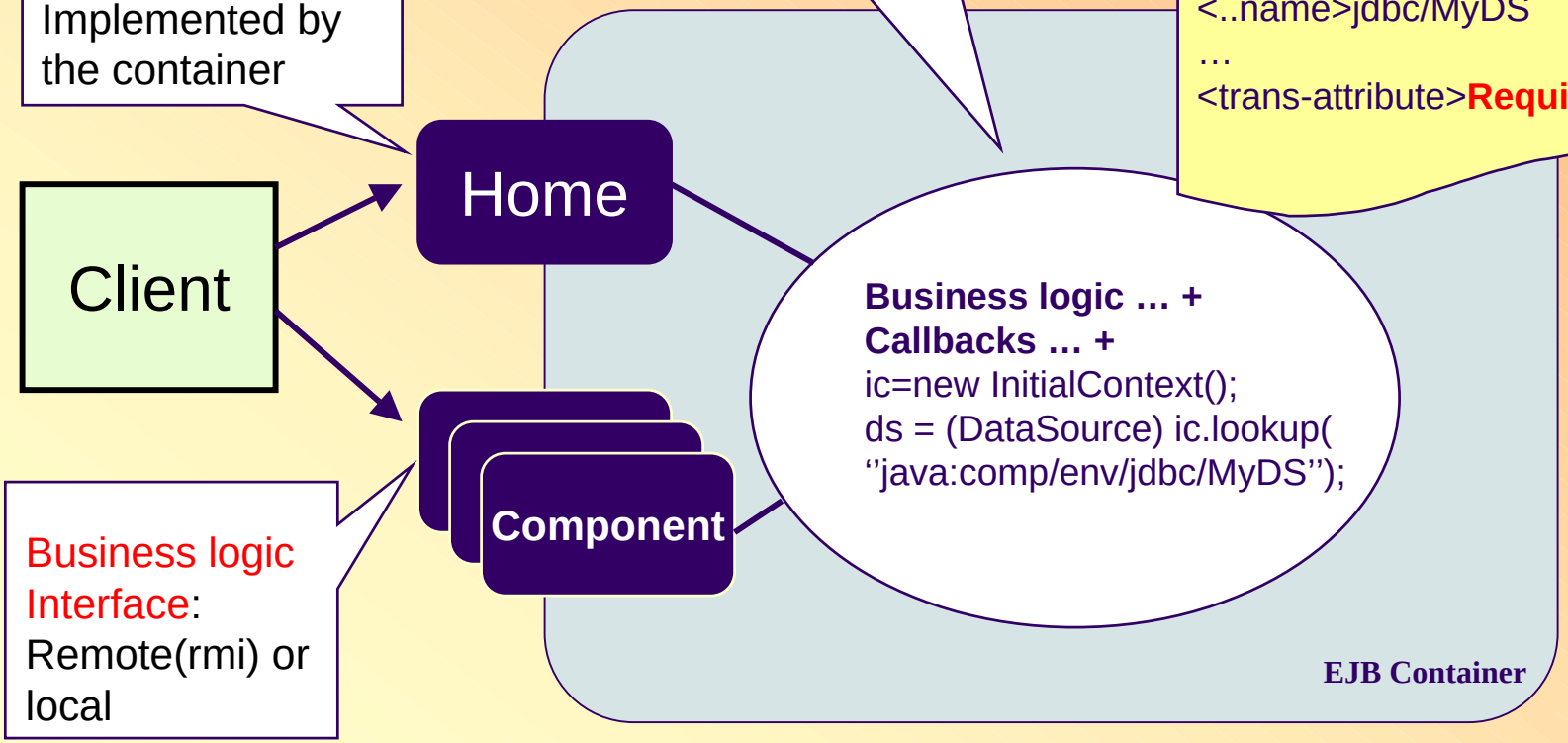
Lifecycle interface
- create
- remove
- find
Remote(rmi) or Local
Implemented by the container

Component implementation:
-business logic code
-container contract callbacks (ejbCreate, ejbPassivate, ejbLoad, …)

Deployment Descriptor:
-transaction behaviour (Tx attributes)
-security definition (ACLs)
-persistence definition (entity beans)
-Resources

DD

```
<resource-ref>
<..name>jdbc/MyDS
…
<trans-attribute>Required
```



Business logic Interface:
Remote(rmi) or local

J2EE™ Development Model

- **Develop presentation and business logic in Java *components***
- **Minimize coding of system related behaviors (transaction, persistence, security, load balancing, pooling, distribution, …), possibly none**
- **Use only logical resources (« ejb/database/mail/JMS/… » factories) through the J2EE™ environment**
- **Use deployment descriptors (xml) to define**
  - Mapping of logical resources to actual resources (assembly)
  - Behaviors such as transactions, persistence (definition + eventually mapping), security
- **Obtain portable, reusable components**

This presenation has ended.
Click anywhere to return.

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